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Assignment #2

Date: _____

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Class: BDS-3

Solution of Question # 1

Simplify the following Boolean function using Three variables maps :-

$$(a) F(x, y, z) = \sum(0, 2, 4, 5)$$

minterms: $0 \rightarrow 000$, $2 \rightarrow 010$, $4 \rightarrow 100$, $5 \rightarrow 101$ K map rows = x , columns = yz in gray code order
(00, 01, 11, 10)

x / yz	00	01	11	10
0	$000 = 0$ 0	$001 = 1$ 0	$011 = 3$ 0	$010 = 2$ 1
1	$100 = 4$ 1	$101 = 5$ 1	$111 = 7$ 0	$110 = 6$ 0

cancel Those that differ

$$\text{Group I } \{0, 4\} = \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z} = \bar{y}\bar{z}$$

$$\text{Group II } \{4, 5\} = x\bar{y}\bar{z} + x\bar{y}z = x\bar{y}$$

$$\text{Group III } \{0, 2\} = \bar{x}\bar{y}\bar{z} + x\bar{y}\bar{z} = \bar{x}\bar{z}$$

Add all groups

$$F(x, y, z) = \bar{y}\bar{z} + x\bar{y} + \bar{x}\bar{z}$$

we pick the minimum set of groups that add
group II and III = $x\bar{y} + \bar{x}\bar{z}$

$$(b) F(x, y, z) = \sum(0, 2, 4, 5, 6)$$

minterms $0 \rightarrow 000$, $2 \rightarrow 010$, $4 \rightarrow 100$, $5 \rightarrow 101$, $6 \rightarrow 110$

K-map using Gray code

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x/yz	00	01	11	10
0	000 = 0	001 = 1	011 = 3	010 = 2
1	100 = 4	101 = 5	111 = 7	110 = 6
	0	0	0	0

$$\text{Group I } (0, 1, 3, 2) = \bar{x}\bar{y}z + \bar{x}\bar{y}z + \bar{x}\bar{y}z = \bar{x}$$

$$\text{Group II } (1, 5) = \bar{x}yz + x\bar{y}z = \bar{y}z$$

$$F(x, y, z) = \bar{x} + \bar{y}z$$

$$(d) F(x, y, z) = \Sigma(1, 2, 3, 7)$$

minterms using 421 $1 \rightarrow 001, 2 \rightarrow 010, 3 \rightarrow 011, 7 \rightarrow 111$

K map using 421 code

x/yz	00	01	11	10
0	000 = 0	001 = 1	011 = 3	010 = 2
1	100 = 4	101 = 5	111 = 7	110 = 6
	0	0	1	0

$$\text{Group 1 } (1, 3, 2) = \bar{x}\bar{y}z + \bar{x}\bar{y}z = \bar{x}z$$

$$\text{Group 2 } (3, 7) = \bar{x}yz + xyz = yz$$

$$\text{Group 3 } (3, 2) = \bar{x}yz + \bar{x}\bar{y}z = \bar{x}y$$

$$F(x, y, z) = \bar{x}z + yz + \bar{x}y$$

Solution of Question #2

Simplify the following boolean expression using three variable K-map

$$(a) F = xy + \bar{x}\bar{y}z + \bar{x}yz$$

$x \backslash yz$	$\frac{00}{yz}$	$\frac{01}{yz}$	$\frac{11}{yz}$	$\frac{10}{yz}$
0 $\bar{x}$	0	1	3	2
1 x	4	5	7	6
	0	0	1	1

$$(0, 2) \text{ Group 1} = \bar{x}\bar{y}z + \bar{x}\bar{y}z = \bar{x}\bar{z}$$

$$(7, 6) \text{ Group 2} = xyz + xy\bar{z} = xy$$

$$F(x, y, z) = xy + \bar{x}\bar{z}$$

$$(b) F = \bar{x}\bar{y} + yz + \bar{x}yz$$

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x/y^2	$\begin{smallmatrix} 00 \\ \bar{y}\bar{z} \end{smallmatrix}$	$\begin{smallmatrix} 01 \\ \bar{y}z \end{smallmatrix}$	$\begin{smallmatrix} 11 \\ yz \end{smallmatrix}$	$\begin{smallmatrix} 10 \\ y\bar{z} \end{smallmatrix}$
$0 \bar{x}$	0	1	3	2
$1 x$	4	5	7	6
	1	0	10	0

$$(0, 1, 3, 2) \text{ Group 1} = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz = \bar{x}$$

$$(3, 7) \text{ Group 2} = xy\bar{z} + xyz = yz$$

$$F(x, y, z) = \bar{x} + yz$$

$$(c) F(x, y, z) = \bar{x}\bar{y} + y\bar{z} + \bar{y}z$$

x/y^2	$\begin{smallmatrix} 00 \\ \bar{y}\bar{z} \end{smallmatrix}$	$\begin{smallmatrix} 01 \\ \bar{y}z \end{smallmatrix}$	$\begin{smallmatrix} 11 \\ yz \end{smallmatrix}$	$\begin{smallmatrix} 10 \\ y\bar{z} \end{smallmatrix}$
$0 \bar{x}$	0	1	3	2
$1 x$	4	5	7	6
	1	0	0	1

$$(0, 2, 4, 6) \text{ Group-1} = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + x\bar{y}\bar{z} + xy\bar{z} = \bar{z}$$

$$(2, 3) \text{ Group-2} = \bar{x}y\bar{z} + \bar{z}y\bar{z} = \bar{x}y$$

$$F(x, y, z) = \bar{z} + \bar{x}y$$

$$(d) F(x, y, z) = xy\bar{z} + x\bar{y}z$$

x/y^2	$\begin{smallmatrix} 00 \\ \bar{y}\bar{z} \end{smallmatrix}$	$\begin{smallmatrix} 01 \\ \bar{y}z \end{smallmatrix}$	$\begin{smallmatrix} 11 \\ yz \end{smallmatrix}$	$\begin{smallmatrix} 10 \\ y\bar{z} \end{smallmatrix}$
$0 \bar{x}$	0	1	3	2
$1 x$	4	5	7	6
	0	1	0	1

$$(3) \text{ Group 1} = \bar{x}y\bar{z}$$

$$(5) \text{ Group 2} = x\bar{y}\bar{z}$$

$$(6) \text{ Group 3} = xy\bar{z}$$

$$F(x, y, z) = \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z}$$

Solution of Question #3

Simplify the following boolean function

using four variable K map

$$(a) F(w, x, y, z) = \Sigma(1, 4, 5, 6, 12, 14, 15)$$

minterms using 8421 $\rightarrow 1 \rightarrow 0001$ $4 \rightarrow 0100$ $5 \rightarrow 0101$

$6 \rightarrow 0110$ $12 \rightarrow 1100$ $14 \rightarrow 1110$

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K map using Gray code

wx/yz	yz 00	yz 01	yz 11	yz 10
$\bar{w}\bar{x}$ 00	0	1	3	2
$\bar{w}x$ 01	4	5	7	6
wx 11	12	13	15	14
$w\bar{x}$ 10	8	9	11	10

$$\text{Group 1 } \{4, 5\} = \bar{w}x\bar{y}z + \bar{w}xy\bar{z} = \bar{w}xz$$

$$\text{Group 2 } \{4, 6, 12, 14\} = \bar{w}x\bar{y}z + \bar{w}xy\bar{z} + w\bar{x}\bar{y}z + w\bar{x}y\bar{z} = x\bar{z}$$

$$\text{Group 3 } \{14, 15\} = w\bar{x}y\bar{z} + wx\bar{y}z = wxy$$

$$\text{Group 4 } \{1, 5\} = \bar{w}\bar{x}\bar{y}z + \bar{w}x\bar{y}z = \bar{w}\bar{y}z$$

Ignore 1 because vert of group include both

$$F(w, x, y, z) = x\bar{z} + wxy + \bar{w}\bar{y}z$$

$$(b) F(A, B, C, D) = \sum(2, 3, 6, 7, 12, 13, 14)$$

minterms using 8421 = 2 $\rightarrow$ 0010 3 $\rightarrow$ 0011 6 $\rightarrow$ 0110 7 $\rightarrow$ 0111

12 $\rightarrow$ 1100 13 $\rightarrow$ 1101 14 $\rightarrow$ 1110

AB/CD	CD 00	$\bar{C}D$ 01	$C\bar{D}$ 11	$\bar{C}\bar{D}$ 10
$\bar{A}\bar{B}$ 00	0	1	3	2
$\bar{A}B$ 01	4	5	7	6
AB 11	12	13	15	14
$A\bar{B}$ 10	8	9	11	10

$$\text{Group 1 } \{3, 2, 7, 6\} = \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} = \bar{A}\bar{C}$$

$$\text{Group 2 } \{12, 13\} = AB\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} = AB\bar{C}$$

$$\text{Group 3 } \{12, 14\} = AB\bar{C}\bar{D} + AB\bar{C}D = AB\bar{D}$$

$$F(A, B, C, D) = \bar{A}\bar{C} + AB\bar{C} + AB\bar{D}$$

$$(c) F(w, x, y, z) = \sum(1, 3, 4, 5, 6, 7, 9, 11, 13, 15)$$

minterms using 8421 1 $\rightarrow$ 0001 3 $\rightarrow$ 0011

5 $\rightarrow$ 0101 6 $\rightarrow$ 0110

15 $\rightarrow$ 1111

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wx/yz	yz 00	yz 01	yz 11	yz 10
$\bar{w}\bar{x}$ 00	0	1	3	2
$\bar{w}x$ 01	4	5	7	6
wx 11	12	13	15	14
$w\bar{x}$ 10	8	9	11	10

Group 1 $\{1, 3, 5, 7, 13, 15, 9, 11\} = \bar{w}\bar{x}yz + \bar{w}xy + \bar{w}yz + wx\bar{y}z + wx\bar{y} + wxy + w\bar{x}yz + w\bar{x}y = 2$

Group 2 $\{4, 6\} = \bar{w}x\bar{y}z + \bar{w}xy = \bar{w}x\bar{z} = \bar{w}x$

$$F = 2 + x\bar{w}$$

(d) $F(A, B, C, D) = \sum \{0, 2, 4, 5, 6, 7, 8, 10, 13, 15\}$

AB/CD	CD 00	CD 01	CD 11	CD 10
AB 00	0	1	3	2
AB 01	4	5	7	6
AB 11	12	13	15	14
AB 10	8	9	11	10

Group 1 $\{5, 7, 13, 15\} = \bar{A}\bar{B}\bar{C}D + \bar{A}BCD + AB\bar{C}D + ABCD = BD$

Group 2 $\{0, 2, 3, 10\} = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}BC\bar{D} + AB\bar{C}\bar{D} + ABC\bar{D} = \bar{B}\bar{D}$

Group 3 $\{4, 6\} = \bar{A}B\bar{C}\bar{D} + \bar{A}BC\bar{D} = \bar{A}B\bar{D}$ ignore D

$$F = BD + \bar{B}\bar{D} + \bar{A}B$$

Solution of Question # 4

Simplify the following expression using four variable Kmap.

(a) $w'z + xz + x'y + wx'z$

wx/yz	yz 00	yz 01	yz 11	yz 10
$\bar{w}\bar{x}$ 00	0	1	3	2
$\bar{w}x$ 01	4	5	7	6
wx 11	12	13	15	14
$w\bar{x}$ 10	8	9	11	10

Group 1 $\{1, 3, 5, 7, 13, 15, 9, 11\} = \bar{w}\bar{x}yz + \bar{w}x\bar{y}z + \bar{w}xy + wx\bar{y}z + wx\bar{y} + wxy + w\bar{x}yz + w\bar{x}y = 2$

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$$\text{Group 2 } (3, 2, 11, 10) = w\bar{x}\bar{y}z + \bar{w}\bar{x}y\bar{z} + w\bar{x}y\bar{z} + \bar{w}\bar{x}\bar{y}z = xy$$

$$F = 2 + \bar{x}y$$

$$(b) F = AB' + B'C'D + BCD' + BCD$$

AB \ CD	00	01	11	10
00 $\bar{A}\bar{B}$	0	1	3	2
01 $\bar{A}B$	4	5	1	6
11 AB	12	13	15	14
10 $A\bar{B}$	8	9	11	10

$$\text{Group 1 } (1, 5, 13, 9) = \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C}D + AB\bar{C}D + A\bar{B}\bar{C}D = \bar{C}D$$

$$\text{Group 2 } (1, 14, 8, 10) = AB\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D} + AB\bar{C}D + A\bar{B}\bar{C}D = A\bar{D}$$

$$\text{Group 3 } (14, 6) = \bar{A}B\bar{C}\bar{D} = \bar{A}B\bar{C}\bar{D} = B\bar{C}\bar{D}$$

$$F = \bar{C}D + A\bar{D} + B\bar{C}\bar{D}$$

$$(c) AB\bar{C} + B'C'D + BCD + A'BC' + A'BCD$$

AB \ CD	00 $\bar{C}\bar{D}$	01 $\bar{C}D$	11 CD	10 $C\bar{D}$
00 $\bar{A}\bar{B}$	0	1	3	2
01 $\bar{A}B$	4	5	7	6
11 AB	12	13	15	14
10 $A\bar{B}$	8	9	11	10

$$\text{Group 1 } (0, 2, 8, 10) = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD = \bar{B}\bar{D}$$

$$\text{Group 2 } (3, 7, 15, 11) = \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + AB\bar{C}D + A\bar{B}\bar{C}D = CD$$

$$\text{Group 3 } (15, 14, 11, 10) = AB\bar{C}D + AB\bar{C}\bar{D} + A\bar{B}CD + A\bar{B}\bar{C}D = AC$$

$$\text{Group 4 } (5, 7) = \bar{A}B\bar{C}D + AB\bar{C}D = \bar{A}BD$$

$$F = B'D + CD + A\bar{B} + ABD$$

$$(d) wxy + xy + w\bar{x}z + w\bar{x}$$

wx \ yz	00 $\bar{y}\bar{z}$	01 $\bar{y}z$	11 yz	10 $y\bar{z}$
00 $\bar{w}\bar{x}$	0	1	3	2
01 $\bar{w}x$	4	5	7	6
11 wx	12	13	15	14
10 $w\bar{x}$	8	9	11	10

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$$F = xw + x^2 + xy$$

Solution of Question # 5

Find the prime implicit of boolean Function are essential.

$$(a) (w, x, y, z) = \Sigma (0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$$

$wx \backslash yz$	00 $\bar{y}\bar{z}$	01 $\bar{y}z$	11 $y\bar{z}$	10 $y\bar{z}$
00 $\bar{w}\bar{x}$	0	1	3	2
01 $\bar{w}x$	4	5	7	6
11 $w\bar{x}$	12	13	15	14
10 wx	8	9	11	10

Possible Groups

Prime Implicants

$$G_1 \{0, 2, 4, 6\}$$

$$G_1 \{5, 7, 13, 15\} = \bar{w}x\bar{y}z + \bar{w}xy\bar{z} + wx\bar{y}z + wx\bar{y}\bar{z} = xz$$

$$G_1 \{4, 5, 7, 6\}$$

$$G_1 \{0, 2, 8, 10\} = \bar{w}\bar{x}\bar{y}z + \bar{w}\bar{x}y\bar{z} + w\bar{x}\bar{y}z + w\bar{x}y\bar{z} = \bar{x}z$$

$$G_1 \{0, 2, 8, 10\}$$

$$G_1 \text{ non essential } \{0, 2, 4, 6\} = \bar{w}\bar{x}\bar{y}z + \bar{x}\bar{w}\bar{y}z + \bar{w}\bar{x}y\bar{z} + \bar{w}x\bar{y}\bar{z} = \bar{w}\bar{z}$$

$$G_1 \{5, 7, 13, 15\}$$

$$G_1 \{4, 5, 7, 6\} = \bar{w}x$$

$$F(A, B, C, D) = \Sigma (0, 2, 3, 5, 7, 8, 10, 11, 14, 15)$$

(b)

$AB \backslash CD$	00 $\bar{C}\bar{D}$	01 $\bar{C}D$	11 CD	10 $C\bar{D}$
00 $\bar{A}\bar{B}$	0	1	3	2
01 $\bar{A}B$	4	5	7	6
11 AB	12	13	15	14
10 $A\bar{B}$	8	9	11	10

Possible Groups

Essentials =

$$\{0, 2, 8, 10\} = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} = \bar{B}\bar{D}$$

$$G_1 \{0, 2, 8, 10\} \rightarrow E$$

$$G_1 \{15, 14, 11, 10\} \rightarrow E \quad G_1 \{5, 7\} = \bar{A}\bar{B}CD + \bar{A}BCD = \bar{A}BD$$

$$G_1 \{3, 7, 15, 11\} \rightarrow N$$

$$G_1 \{15, 14, 11, 10\} = ABCD + ABC\bar{D} + \bar{A}B\bar{C}D + \bar{A}B\bar{C}\bar{D} = AD$$

$$G_1 \{5, 7\} \rightarrow E$$

non Essentials = $CD, B'C$

$$G_1 \{3, 7, 11, 10\} \rightarrow N$$

$$F = \bar{B}\bar{D} + \bar{A}BC + AC + (CD \text{ or } B'C)$$

$$(C) F(A, B, C, D) = \Sigma \{2, 3, 4, 5, 6, 7, 9, 11, 12, 13\}$$

$AB \backslash CD$	00 $\bar{C}\bar{D}$	01 $\bar{C}D$	11 CD	10 $C\bar{D}$
00 $\bar{A}\bar{B}$	0	1	3	2
01 $\bar{A}B$	4	5	7	6
11 AB	12	13	15	14
10 $A\bar{B}$	8	9	11	10

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01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$\text{Essential} = x'z, wyz, xyz$$

$$F = xz + wyz + xyz$$

Solution of Question #6

Simplify the following boolean function F to get it with the don't care condition d and express the simplified function in sum of min terms form.

$$a) F(x, y, z) = \sum (0, 1, 4, 5, 6) \quad d(x, y, z) = \sum (2, 3, 7)$$

$x \backslash yz$	00	01	11	10
0	0	1	3	2
1	4	5	7	6

$$F = 1$$

$$F = \sum (0, 1, 2, 3, 4, 5, 6, 7)$$

$$b) (A, B, C, D) = \sum (0, 6, 8, 13, 14) \quad d(A, B, C, D) = \sum (2, 11, 10)$$

$AB \backslash CD$	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$c) F = A'D + B'D + BCD + ABCD \quad \text{or} \quad F = \sum (0, 2, 4, 6, 8, 10, 13, 14)$$

$$\downarrow F(A, B, C, D) = \sum (5, 6, 7, 12, 14, 15) \quad d(A, B, C, D) = \sum (3, 9, 11, 15)$$

$AB \backslash CD$	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

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$$F = BC + CD + ABD' + ABD \text{ and } F = 4 \quad F = \Sigma(3, 5, 6, 7, 11, 12, 14, 15)$$

$$(d) F(A, B, C, D) = \Sigma(4, 12, 7, 2, 10) \text{ and } (A, B, C, D) = \Sigma(0, 6, 8)$$

AB \ CD	00	01	11	10
00	0 X	1	3	2 1
01	4 1	5	7 1	6 X
11	12 1	13	15	14
10	8 X	9	11	10 1

Solution of Question #7.

Write the following boolean expression in product of forms.

$$a'b + a'c' + abc$$

a	b	c	ab	abc	F
0	0	0	0	0	1 0
0	0	1	0	0	0 1
0	1	0	1	0	1 2
0	1	1	1	0	1 3
1	0	0	0	0	0 4
1	0	1	0	0	0 5
1	1	0	0	0	0 6
1	1	1	0	1	1 7

Product of sums:

a \ bc	00	01	11	10
0	0	1 0	3	2
1	4 0	5 0	7	6 0

$$F' = bc + ac'$$

$$F'' = F(b'c + ac')'$$

$$(b'c) \cdot (ac')$$

$$F = [b + c'] \cdot [a' + c]$$

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Solution of Question # 8

convert each of following to other canonical form.

(a) $F(x, y, z) = \sum (1, 3, 5)$

Three variables = 8 minterms 0, 1, 2, 3, 4, 5, 6, 7

Given minterm, 1, 3, 5 so max term are remaining
0, 2, 4, 6, 7

So $F(x, y, z) = \prod (0, 2, 4, 6, 7)$

(b) $F(A, B, C, D) = \prod (3, 5, 8, 11)$

4 variable = 16 minterm (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15)

Given minterm (3, 5, 8, 11) so min are remaining

$F(A, B, C, D) = \sum (1, 2, 4, 6, 7, 9, 10, 12, 13, 14, 15)$

Solution of Question # 9

$F = xy'z + x'y'z + wxy + wx'y + wxz$

(a) obtained the truth table for F.

w	x	y	z	w'	x'	y'	xy'z	x'y'z	wxy	F
0	0	0	0	1	1	1	0	0	0	0
0	0	0	1	1	1	1	0	1	0	1
0	0	1	0	1	1	0	0	0	0	0
0	0	1	1	1	1	0	0	0	0	0
1	1	0	0	1	0	1	0	0	0	0
0	1	0	1	1	0	1	1	0	0	1
0	1	1	0	1	0	0	0	0	0	1
0	1	1	1	1	0	0	0	0	0	1
1	0	0	0	0	1	1	0	0	0	0
1	0	0	1	0	1	1	0	1	0	1
1	0	1	0	0	1	0	0	0	1	1
1	0	1	1	0	1	0	0	0	1	1
1	1	0	0	0	0	1	0	0	0	0
1	1	0	1	0	0	1	1	0	0	1
1	1	1	0	0	0	0	0	0	0	1
1	1	1	1	0	0	0	0	0	0	1

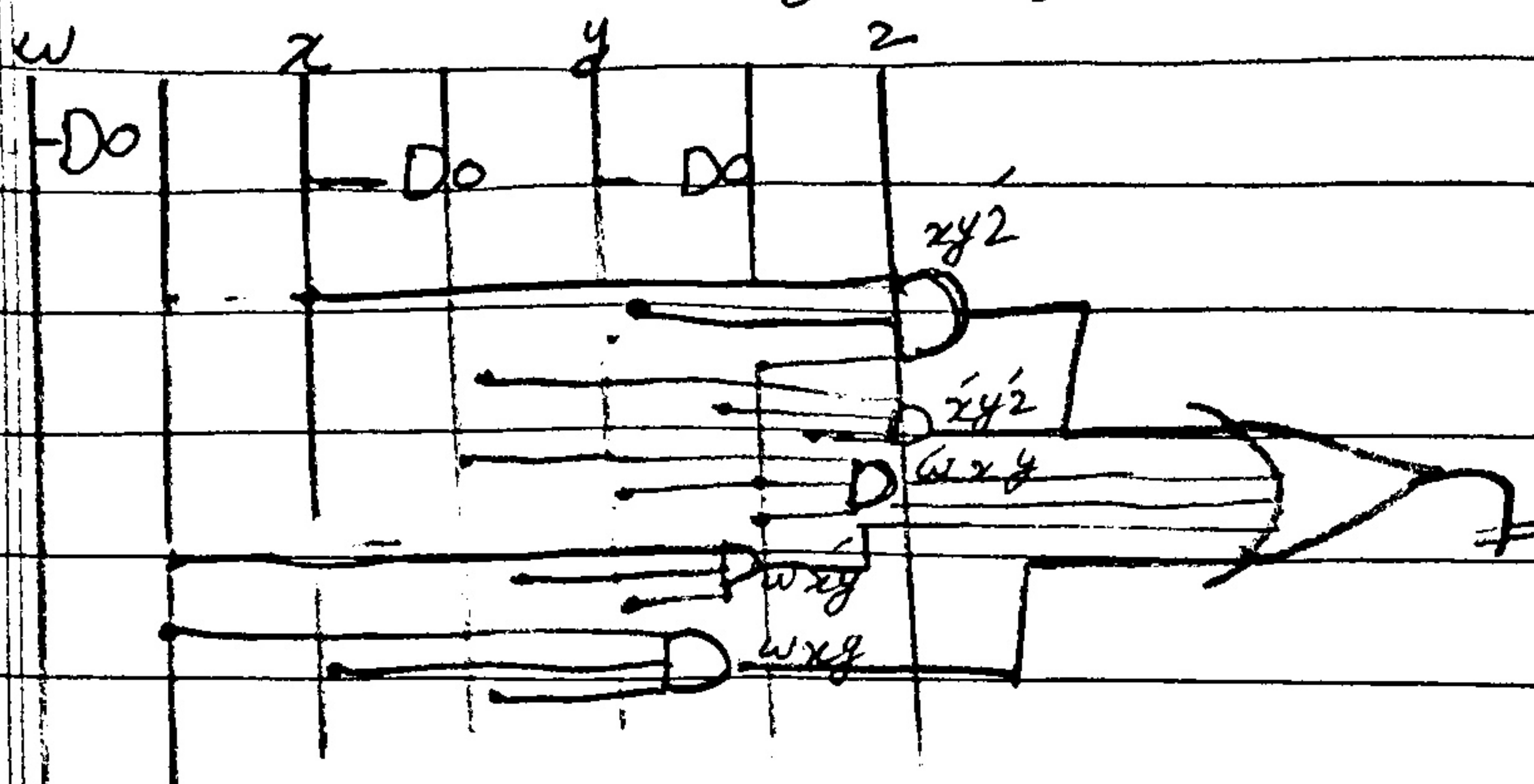
$F = \sum (1, 5, 6, 7, 9, 10, 11, 13, 14, 15)$

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(b) Draw the logic diagram using the original boolean expression

$$x y' z + x' y z + w x y + w x y + w x y$$



(c) use boolean algebra to simplify the functions in a minimum number of literals.

$$\begin{aligned} F &= x y' z + x' y z + w x y + w x y + w x y + w x y \\ &= y' z (x + x') + x y (w + w) + w y (x' + x) \\ &= y' z + x y + w y \\ &= y' z + y (x + w) \end{aligned}$$

(d) obtain truth table of simplified version

w	x	y	z	w'	y'	z	y'z	wx'y(w'z)	F
0	0	0	0	1	1	1	0	0	0
0	0	0	1	1	1	0	1	0	1
0	0	1	0	1	0	1	0	0	0
0	0	1	1	1	0	0	0	0	0
0	1	0	0	1	1	1	0	1	0
0	1	0	1	1	1	0	1	1	1
0	1	1	0	1	0	1	0	1	1
0	1	1	1	1	0	0	0	1	1

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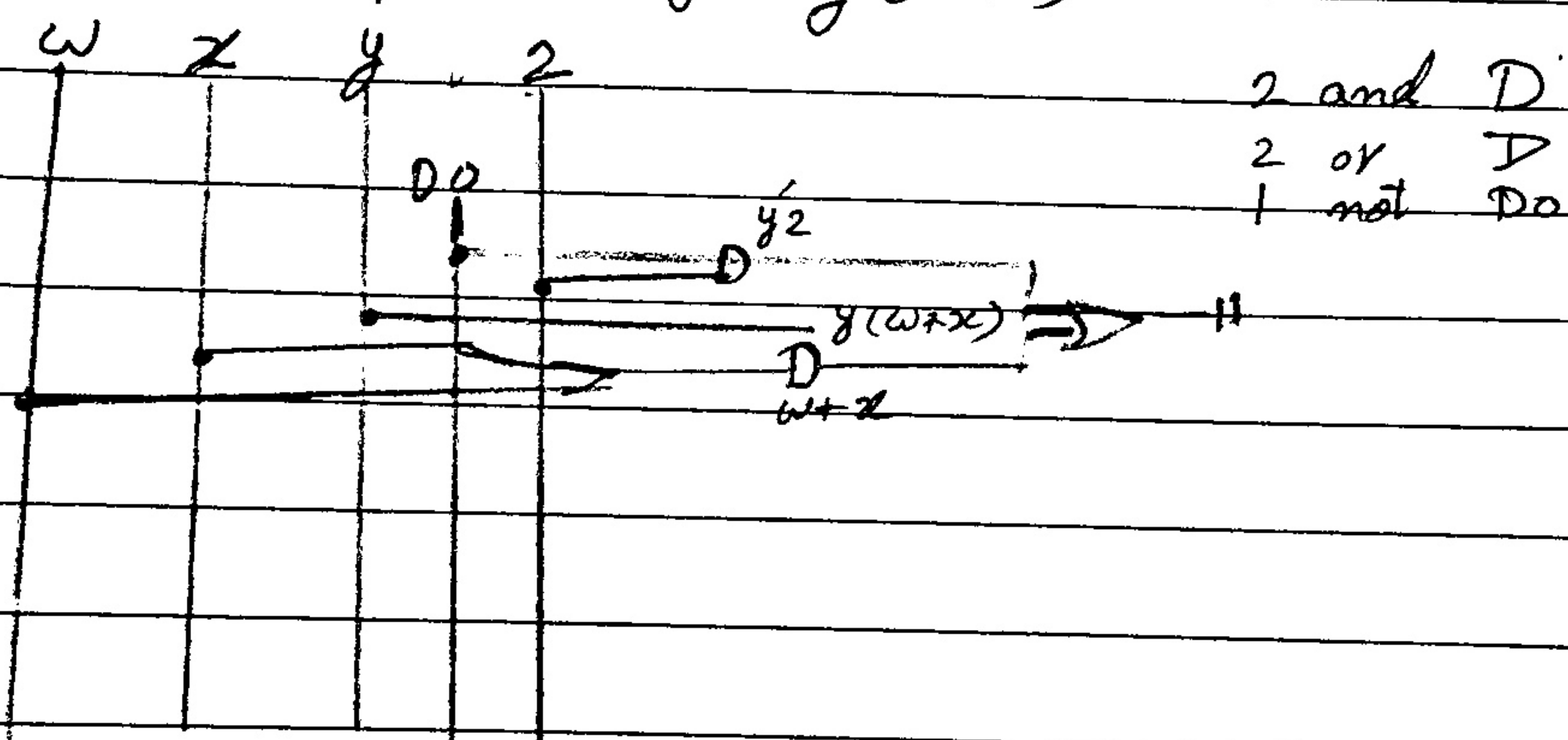
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w	x	y	z	w	y'	z	y'z	wtx	y(wtx)	F
1	0	0	0	0	1	1	0	1	0	1
1	0	0	1	0	1	0	1	1	0	0
1	0	1	0	0	1	1	0	1	1	1
1	0	1	1	0	1	0	0	1	1	1
1	1	0	0	0	0	1	0	1	0	1
1	1	0	1	0	0	0	1	1	0	1
1	1	1	0	0	0	1	0	1	1	1
1	1	1	1	0	0	0	0	1	1	1

$$F = \sum (1, 5, 6, 7, 9, 10, 11, 13, 14, 15)$$

(c) Draw the logic diagram from simplified expression him

$$F(\text{simplified}) = y'z + y(w+x)$$



Solution of Question # 10

Simplify the following Boolean function T_1 and T_2 to minimum number of literals.

A	B	C	T_1	T_2
0	0	0	1	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	0	1

Day: _____

Date: _____

$$T_1 = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}B\bar{C} \quad \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}B\bar{C} + \bar{A}B\bar{C}$$

$$T_2 = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} \quad \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C}$$

$$T_1 = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} \quad \text{added}$$

$$= \bar{A}\bar{B}(\bar{C} + \bar{C}) + \bar{A}\bar{C}(B + \bar{B})$$

$$= \bar{A}\bar{B} + \bar{A}\bar{C}$$

$$= \bar{A}(\bar{B} + \bar{C})$$

$$T_2 = \bar{A}\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} \quad \text{added}$$

$$= \bar{B}C(\bar{A} + \bar{A}) + \bar{A}\bar{B}(\bar{C} + \bar{C}) + \bar{A}\bar{B}(\bar{C} + \bar{C})$$

$$= \bar{B}C + \bar{A}\bar{B} + \bar{A}\bar{B}$$

$$= \bar{B}C + \bar{A}(\bar{B} + \bar{B})$$

$$= \bar{B}C + \bar{A}$$